

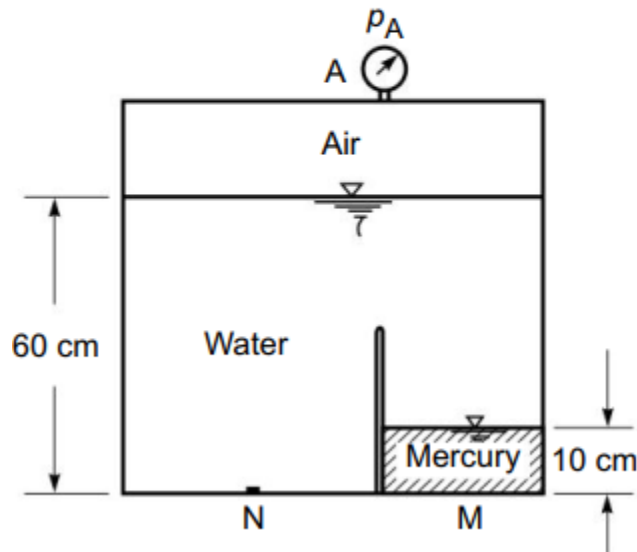
Topic 2: Fluid pressure and its measurement

1.



- 7 For the system shown in Fig. 2.77 calculate the air pressure p_A to make the pressure at N one third of that at M.

(Ans. $p_A = 0.2937$ kPa)



2.



In construction, a barometer is a graduated inverted tube with its open end dipped in the measuring liquid contained in a trough opened to atmosphere.

Estimate the height of liquid column in the barometer where the atmospheric pressure is 100 kN/m^2 . (a) when the liquid is mercury and (b) when the liquid is water. The measuring temperature is 50°C , the vapour pressures of mercury and water at this temperature are respectively $0.015 \times 10^4 \text{ N/m}^2$ and $1.23 \times 10^4 \text{ N/m}^2$, and the densities are 13500 and 980 kg/m^3 respectively. What would be the percentage error if the effect of vapour pressure is neglected.

Ans. (0.754 m, 9.12 m, 0.14%, 14.05%)

3. 

Two pipes A and B (Fig. 2.41) are in the same elevation. Water is contained in A and rises to a level of 1.8 m above it. Carbon tetrachloride (Sp. gr. = 1.59) is contained in B . The inverted U-tube is filled with compressed air at 300 kN/m^2 and 30°C . Barometer reads 760 mm of mercury. Determine:

- The pressure difference in kN/m^2 between A and B if $z = 0.45 \text{ m}$.
- The absolute pressure in B in mm of mercury.

Ans. ($P_B - P_A = 3.4 \text{ kN/m}^2$, 2408.26 mm)

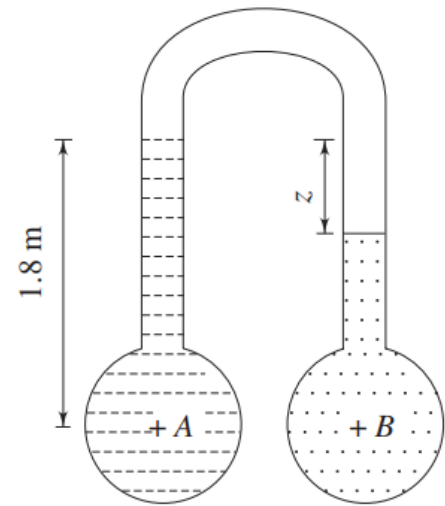
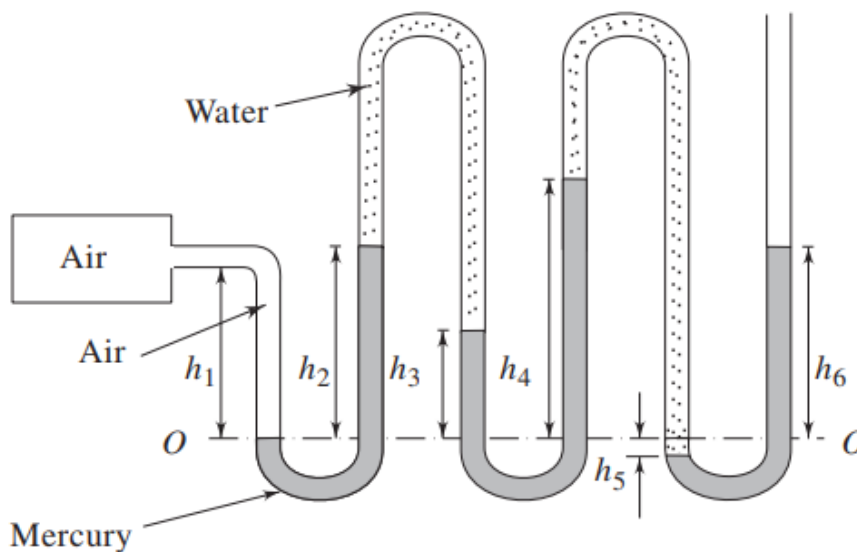


Fig. 2.41 Pipes with water and carbon tetrachloride

4. 

A multi-tube manometer using water and mercury is used to measure the pressure of air in a vessel, as shown in Fig. 2.42. For the given values of heights, calculate the gauge pressure in the vessel. $h_1 = 0.4 \text{ m}$, $h_2 = 0.5 \text{ m}$, $h_3 = 0.3 \text{ m}$, $h_4 = 0.7 \text{ m}$, $h_5 = 0.1 \text{ m}$ and $h_6 = 0.5 \text{ m}$.

Ans. (190.31 kN/m^2 gauge)



5. 

For the system shown in Fig. 2.78 determine the pressures at A and B.

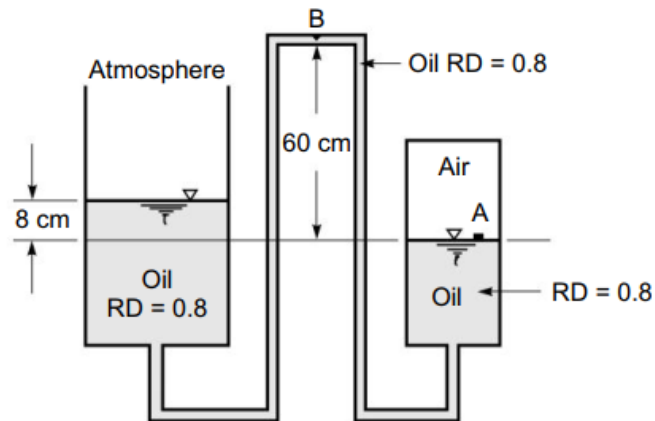


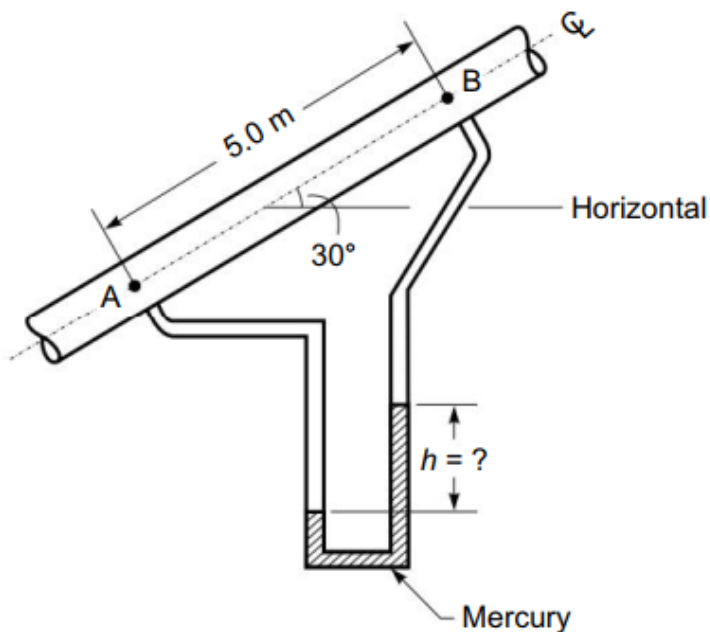
Fig. 2.78 Problem 2.8

(Ans. $p_A = 626.6$ Pa (gauge),
 $p_B = -4072.6$ Pa (gauge))

6. 

If the pipe in Fig. 2.79 contains water and there is no flow, calculate the value of the manometer reading h .

(Ans. $h = 0$)



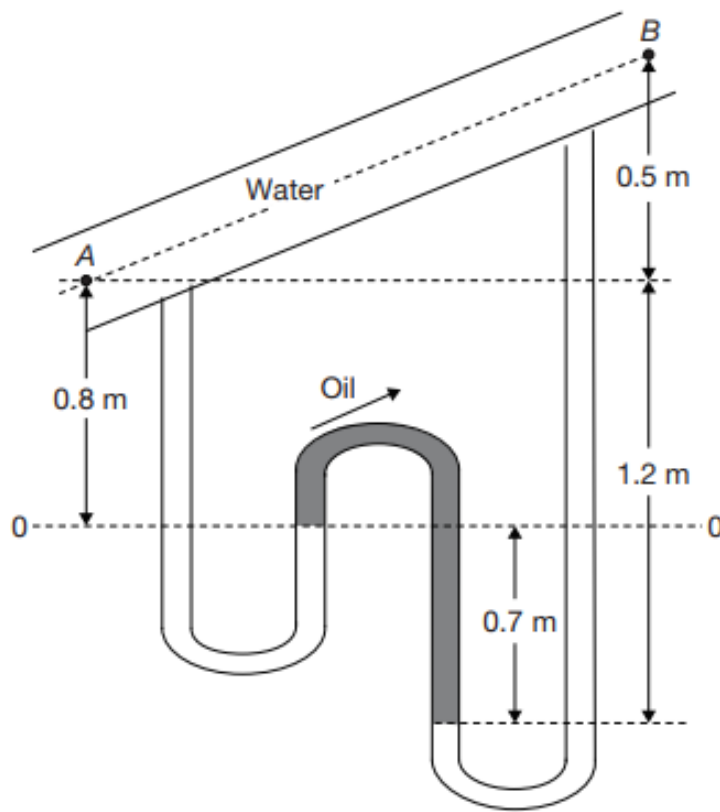
7. 

A certain fluid of specific gravity 0.8 flows upwards through a vertical pipe. A and B are two points on the pipe, B being 0.3 m higher than A. A U-tube mercury manometer is connected at points A and B. If the difference in pressure between A and B is 5 kPa, find the difference in the heights of the mercury columns in the manometer.

(Ans. $h = 21.1 \text{ mm}$)

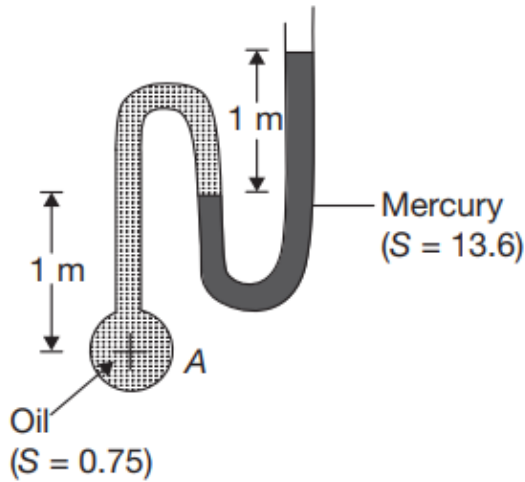
8. 

An inverted U-tube differential manometer is used to measure pressure difference in an inclined water pipe as shown in the figure. The manometer fluid is oil, of specific gravity 0.75



Ans. 3678.75 N/m^2

9.



For the U-tube arrangement shown in the figure, pressure at A (in kN/m^2) is _____.

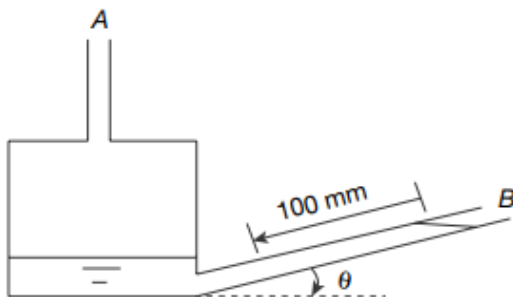
- (A) 140.23 (B) 140.77
(C) 140.98 (D) 140.62

Ans. B

10. GATE 2004



In the inclined manometre shown in the figure below, the reservoir is large. Its surface may be assumed to remain at a fixed elevation. A is connected to a gas pipeline and the deflection noted on the inclined glass tube is 100 mm. Assuming ($\theta = 30^\circ$ and the manometric fluid as oil with specific gravity of 0.86, the pressure at A is [2004]



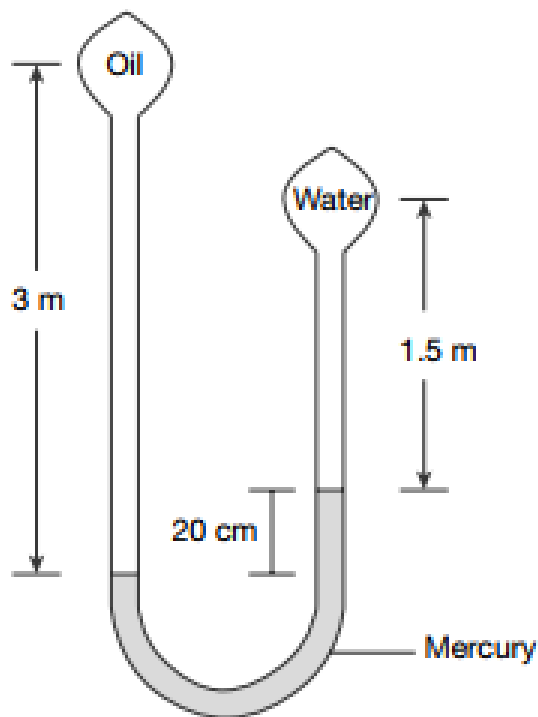
- (a) 43 mm water (vacuum)
(b) 43 mm water
(c) 86 mm water
(d) 100 mm water

Ans. B

11. GATE 2003



Two pipelines, one carrying oil mass density 900 kg/m^3 and the other water, are connected to a manometre as shown in the figure. By what amount the pressure in the water pipe should be increased, so that the mercury levels in both the limbs of the manometre become equal? Mass density of mercury = $13,550 \text{ kg/m}^3$ and $g = 9.81 \text{ m/s}^2$. [2003]



- | | |
|--------------|--------------|
| (a) 24.7 kPa | (b) 26.5 kPa |
| (c) 26.7 kPa | (d) 28.9 kPa |

Ans. A